Animal and Object Detection using YOLOv8

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Tier 1

The Problem

- Problem: Identifying animal species from images can be time-consuming and requires expert knowledge.
- Who cares: Wildlife researchers, conservationists, and biodiversity organizations.
- Why it’s important: Automating animal and object recognition helps speed up wildlife monitoring and supports biodiversity research efforts.

My Solution

- A YOLOv8 object detection system that analyzes uploaded images and identifies detected objects using bounding boxes and confidence scores.

- How it works:

Image → YOLOv8 Model → Object Detection Results

Technical Approach

- Computer Vision Technique: Object Detection
- Model: YOLOv8 (pre-trained)
- Framework: Ultralytics YOLO + Python

Justification:

Pre-trained YOLOv8 models provide fast and effective object detection for real-world image testing and analysis.

Data Plan

- Source: Personal test images and publicly available animal images
- Content: Real dogs, backyard cat, stuffed animal objects, decorative objects, and wildlife images
- Purpose: Evaluate strengths and limitations of pretrained YOLOv8 object detection

System Diagram

- [Input Image] → [YOLOv8 Model] → [Detection Results]

Initial Testing

- Initial testing was performed using a variety of publicly available images to confirm the YOLOv8 model was functioning correctly before focused testing began.

```
[2]: (np.float64(-0.5), np.float64(399.5), np.float64(399.5), np.float64(-0.5))
```

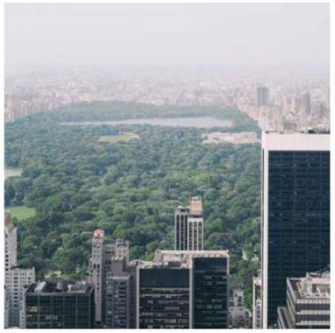
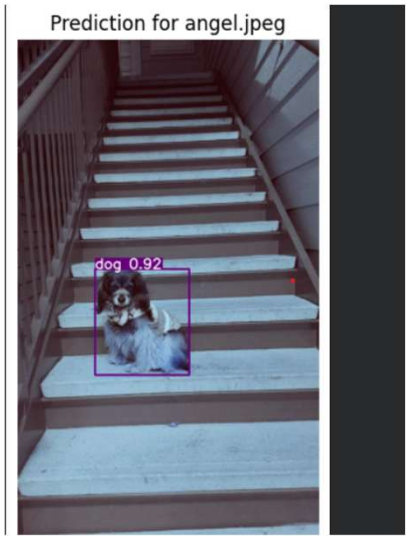


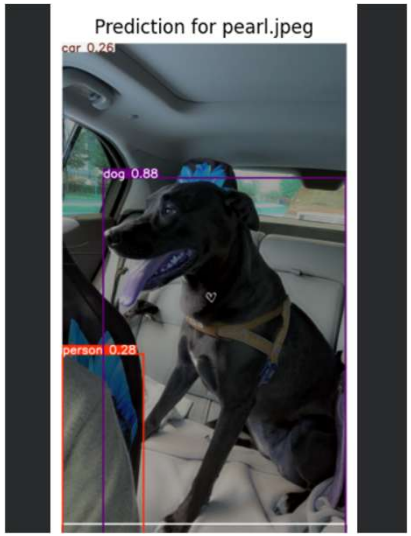
Image 2

Label: peacock

Successful Detections

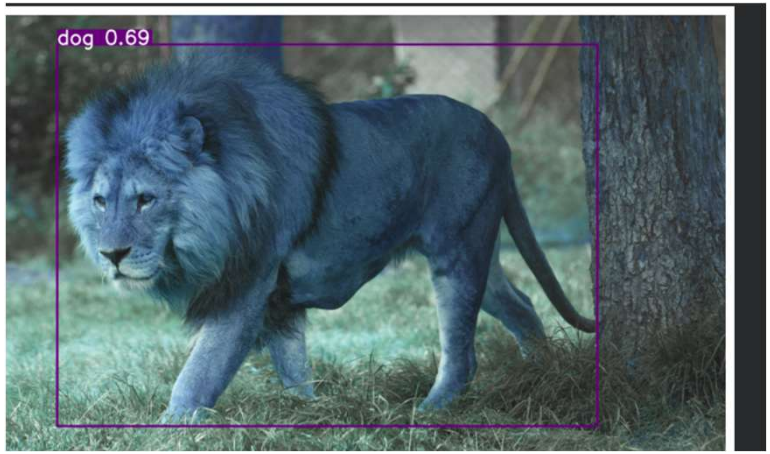
- YOLOv8 successfully detected real dogs in multiple test images

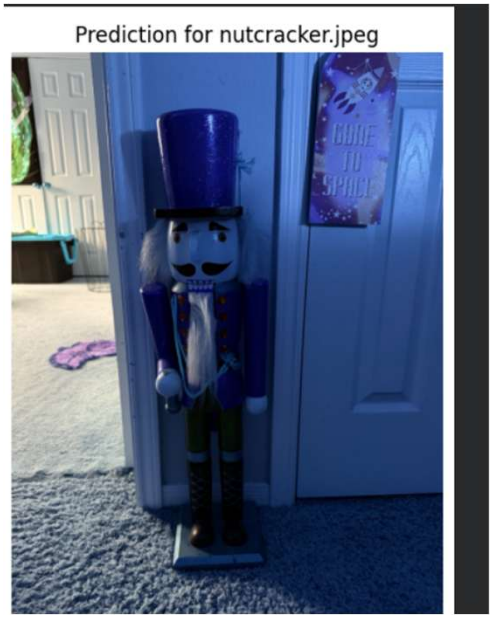




Incorrect/Failed Detections

The model showed limitations with species-level accuracy and unusual objects.





Observations and Findings

- YOLOv8 successfully detected real dogs in multiple test images.
- The model struggled with species-level accuracy in some cases.
- A lion image was incorrectly classified as a dog.
- Decorative and unusual objects were more difficult for the model to detect.
- Pretrained object detection models perform well on common objects but may struggle with ambiguous inputs.

Success Metrics

Metric Type	Metric	Target
Primary	Successful object detection and analysis of model performance	Correctly identify common objects/animals in most test cases
Secondary	Inference Time	< 1 second per image

Challenges & Backup Plans

- Challenge: Incorrect or failed detections
- Plan B: Analyze model limitations and compare successful vs unsuccessful predictions
- Challenge: Limited training/customization time
- Plan B: Use pretrained YOLOv8 model for real-world testing and evaluation

Resources Needed

Resource	Personal and publicly available test images; GitHub repository
Compute	Google Colab (free GPU)
Frameworks	Ultralytics YOLOv8; Python
Estimated Cost	$0